

Network Protocols Assignment

Task

Fill the table below with information about **top 100 network protocols**.

Table

Protocol	Layer	Port Number	TCP/UDP	Encryption	Usage
HTTP	Application	80	TCP	No	Web browsing
HTTPS	Application	443	TCP	Yes - TLS	Secure Web browsing
SSH	Application	22	TCP	Yes	Secure remote administration
FTP	Application	20, 21	TCP	No	File transfer (data, control)
SFTP	Application	22	TCP	Yes	Secure file transfer over SSH
Telnet	Application	23	TCP	No	Unsecure remote terminal access
FTPS	Application	990,989	TCP	Yes - TLS	FTP with SSL/TLS encryption
SMTP	Application	25	TCP	Optional - StartTLS	Outgoing email between servers
POP3	Application	110	TCP	Optional - StartTLS	Downloading email from a server
IMAP4	Application	143	TCP	Optional – StartTLS	Managing emails on a server
IMAPS	Application	993	TCP	Yes - TLS	Secure IMAP connection
POP3S	Application	995	TCP	Yes - TLS	Secure POP3 connection
DNS	Application	53	TCP/UDP	No	Domain names to IP
DHCP	Application	67,68	UDP	No	Automatically assigning Ips
NTP	Application	123	UDP	No	Synchronize system over a network
SNMP	Application	161, 162	UDP	No	Network device monitoring
LDAP	Application	389	TCP/UDP	Optional	Directory services
LDAPS	Application	636	TCP	Yes - TLS	Secure LDAP
RDP	Application	3389	TCP	Yes	Remote desktop access
SMB/CIFS	Application	445, 139	TCP	Yes	File and printer sharing
SIP	Application	5060	TCP/UDP	Optional	VoIP Signaling
SIP - TLS	Application	5061	TCP	Yes	Secure VoIP
RTP	Application	Dynamically – (16384-32767)	UDP	Optional	Voice/Video use even number ports
RTCP	Application	Dynamically	UDP	Optional	RTP control use RTP followed higher odd number port
NETBIOS	Application	137, 139	TCP/UDP	No	Windows network
NFS	Application	2049	TCP/UDP	Optional	File Sharing

AFP	Application	548	TCP	Optional	Apple sharing
MQTT	Application	1883	TCP	Optional	IoT
MQTT-TLS	Application	8883	TCP	Yes	Secure IoT
AMQP	Application	5672	TCP	Optional	Messaging
CoAP	Application	5683	UDP	Optional	IoT
CoAP-DTLS	Application	5684	UDP	Yes	Secure IoT
XMPP	Application	5222	TCP	Optional	Chat
IRC	Application	6667	TCP	Optional	Chat
OSPF	Network	89	IP	Optional	Routing
EIGRP	Network	88	IP	Optional	Routing
ICMP	Network	-	IP	No	Ping
IGMP	Network	-	IP	No	Multicast
IP	Network	-	-	No	Internet layer
IPv6	Network	-	-	Optional	Internet
IPsec	Network	500/4500	UDP	Yes	VPN
L2TP	Network	1701	UDP	Optional	VPN
PPTP	Network	1723	TCP	Weak	VPN
OpenVPN	Application	1194	TCP/UDP	Yes	VPN
WireGuard	Network	51820	UDP	Yes	VPN
ARP	Data-Link	-	-	No	MAC lookup
PPP	Data-Link	-	-	Optional	Dial-up
PPPoE	Data-Link	-	-	Optional	DSL
Ethernet	Data-Link	-	-	No	LAN
VLAN	Data-Link	-	-	No	Virtual LAN
STP	Data-Link	-	-	No	Loop prevention
LACP	Data-Link	-	-	No	Link aggregation
LLDP	Data-Link	-	-	No	Device discovery
CDP	Data-Link	-	-	No	Cisco discovery
Kerberos	Application	88	TCP/UDP	Yes	Auth
Radius	Application	1812	UDP	Optional	Auth
TACACS+	Application	49	TCP	Yes	Auth
Syslog	Application	514	UDP	Optional	Logs
Syslog-TLS	Application	6514	TCP	Yes	Secure logs
Redis	Application	6379	TCP	Optional	Cache
MySQL	Application	3306	TCP	Optional	Database
PostgreSQL	Application	5432	TCP	Optional	Database
MongoDB	Application	27017	TCP	Optional	Database
VNC	Application	5900	TCP	Optional	Remote GUI
TeamViewer	Application	5938	TCP/UDP	Yes	Remote
gRPC	Application	443	TCP	Yes	API
WebSocket	Application	80/443	TCP	Optional	Realtime
QUIC	Transport	443	UDP	Yes	HTTP/3
SCTP	Transport	—	SCTP	Optional	Telecom
TCP	Transport	—	TCP	Optional	Transport
UDP	Transport	—	UDP	No	Transport
DCCP	Transport	—	DCCP	Optional	Streaming
Fibre Channel	Data-Link	—	—	Optional	Storage

iSCSI	Application	3260	TCP	Optional	Storage
SLP	Application	427	TCP/UDP	Optional	Discovery
SSDP	Application	1900	UDP	No	UPnP
UPnP	Application	1900	UDP	Optional	Device
H.323	Application	1720	TCP	Optional	VoIP
MGCP	Application	2427	UDP	Optional	VoIP
GTP	Network	2123	UDP	Optional	Mobile
Diameter	Application	3868	TCP	Yes	Auth
Modbus	Application	502	TCP	No	Industrial
BACnet	Application	47808	UDP	Optional	Building
BGP	Application	179	TCP	Optional	Inter-AS routing
RIP	Application	520	UDP	No	Routing protocol
RIPng	Application	521	UDP	No	IPv6 routing
VXLAN	Network	4789	UDP	Optional	Overlay network
Geneve	Network	6081	UDP	Optional	Virtual networking
CAPWAP	Application	5246	UDP	Optional DTLS	Wireless AP control
TFTP	Application	69	UDP	No	Simple file transfer
Gopher	Application	70	TCP	No	Document retrieval
Finger	Application	79	TCP	No	User info
WHOIS	Application	43	TCP	No	Domain lookup
NNTP	Application	119	TCP	Optional TLS	Usenet
IPP	Application	631	TCP	Optional TLS	Printing
LPD	Application	515	TCP	No	Printer service
RTSP	Application	554	TCP/UDP	Optional TLS	Streaming control
HLS	Application	80/443	TCP	TLS	Video streaming
SMTPS	Application	465	TCP	TLS	Secure mail send
NAT-T	Network	4500	UDP	IPsec	NAT traversal

Instructions

- Fill all columns
 - Keep answers short
 - Write correct port numbers and layers
 - Submit in Word or PDF
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